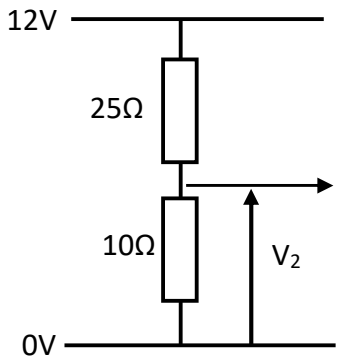
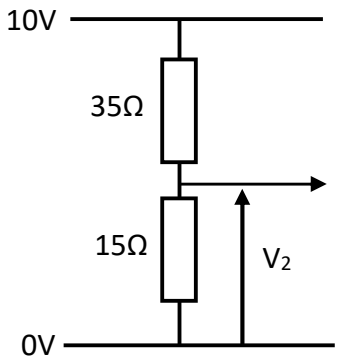
N5 Analogue Electronics – Voltage Divider Worksheet

In your jotter, draw the following circuits and calculate the missing value.
You can verify your answers with Yenka.

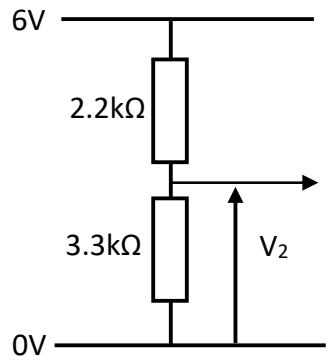
Q1



Q2



Q3



Q1)

$$V_2 = \frac{R_2}{R_1 + R_2} \times V_{Supply}$$

$$V_2 = \frac{10}{10 + 25} \times 12$$

V_2 = **3.43V**

Q2)

$$V_2 = \frac{R_2}{R_1 + R_2} \times V_{Supply}$$

$$V_2 = \frac{15}{15 + 35} \times 10$$

V_2 = **3V**

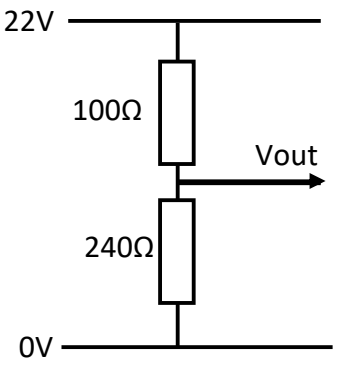
Q3)

$$V_2 = \frac{R_2}{R_1 + R_2} \times V_{Supply}$$

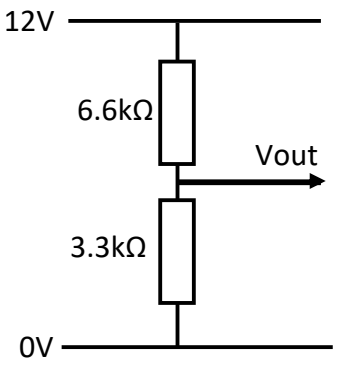
$$V_2 = \frac{3300}{2200 + 3300} \times 6$$

V_2 = **3.6V**

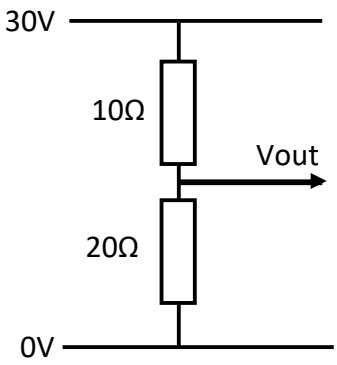
Q4



Q5



Q6



Q4)

$$V_{out} = \frac{R_2}{R_1 + R_2} \times V_{Supply}$$

$$V_{out} = \frac{240}{100 + 240} \times 22$$

V_{out} **= 15.53V**

Q5)

$$V_{out} = \frac{R_2}{R_1 + R_2} \times V_{Supply}$$

$$V_{out} = \frac{3300}{6600 + 3300} \times 12$$

V_{out} **= 4V**

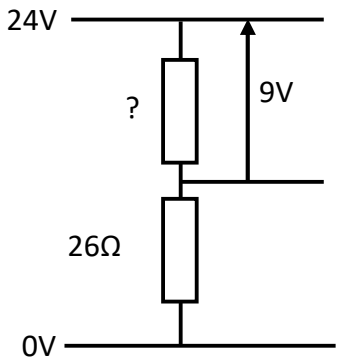
Q6)

$$V_{out} = \frac{R_2}{R_1 + R_2} \times V_{Supply}$$

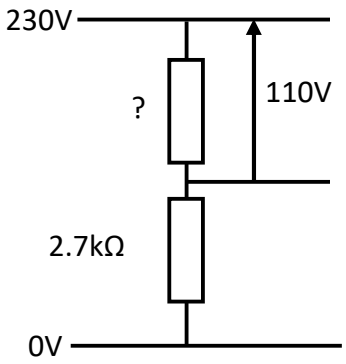
$$V_{out} = \frac{20}{10 + 20} \times 30$$

V_{out} **= 20V**

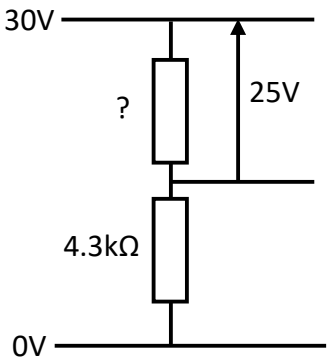
Q7



Q8



Q9



Q7)

$V_{total} = V_1 + V_2$

$V_2 = V_{total} - V_1$

$V_2 = 24 - 9 = 15V$

$$\frac{V_1}{V_2} = \frac{R_1}{R_2}$$

$$R_1 = \frac{V_1}{V_2} \times R_2$$

$$R_1 = \frac{9}{15} \times 26 = \mathbf{15.6\Omega}$$

Q8)

$V_{total} = V_1 + V_2$

$V_2 = V_{total} - V_1$

$V_2 = 230 - 110 = 120V$

$$\frac{V_1}{V_2} = \frac{R_1}{R_2}$$

$$R_1 = \frac{V_1}{V_2} \times R_2$$

$$R_1 = \frac{110}{120} \times 2700 = 2475 = \mathbf{2.5k\Omega}$$

Q9)

$V_{total} = V_1 + V_2$

$V_2 = V_{total} - V_1$

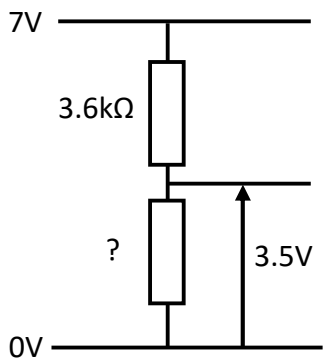
$V_2 = 30 - 25 = 5V$

$$\frac{V_1}{V_2} = \frac{R_1}{R_2}$$

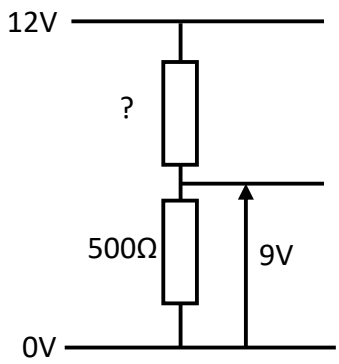
$$R_1 = \frac{V_1}{V_2} \times R_2$$

$$R_1 = \frac{25}{5} \times 4300 = 21500 = \mathbf{21.5k\Omega}$$

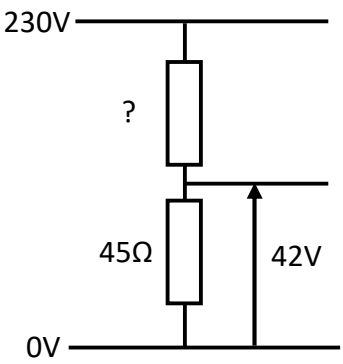
Q10



Q11



Q12



Q10)

$V_{total} = V_1 + V_2$
$V_1 = V_{total} - V_2$
$V_1 = 7 - 3.5 = 3.5V$

$$\frac{V_1}{V_2} = \frac{R_1}{R_2}$$
$$R_2 = \frac{V_2}{V_1} \times R_1$$
$$R_2 = \frac{3.5}{3.5} \times 3600 = 3600 = \mathbf{3.6k\Omega}$$

Q11)

$V_{total} = V_1 + V_2$
$V_2 = V_{total} - V_1$
$V_2 = 12 - 9 = 3V$

$$\frac{V_1}{V_2} = \frac{R_1}{R_2}$$
$$R_1 = \frac{V_1}{V_2} \times R_2$$
$$R_1 = \frac{3}{9} \times 500 = 166.66667 = \mathbf{167k\Omega}$$

Q12)

$V_{total} = V_1 + V_2$
$V_2 = V_{total} - V_1$
$V_2 = 230 - 42 = 188V$

$$\frac{V_1}{V_2} = \frac{R_1}{R_2}$$
$$R_1 = \frac{V_1}{V_2} \times R_2$$
$$R_1 = \frac{188}{42} \times 45 = 201.42857 = \mathbf{201\Omega}$$